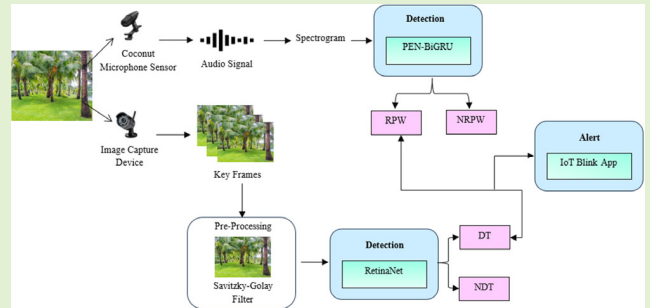


COMICS: Coconut Microphone Sensor and Stem Image-Based IOT Automation in Red Palm Weevil Infestation Detection

L. K. Sowmya Sundari^{ID}, Mallikarjun M. Kodabagi, B. Muthu Kumar, Mohammed Altaf Ahmed, BalaAnand Muthu, and A. Ahilan

Abstract—A smart agriculture system integrates sensors into coconut trees (CoTs) to collect essential data for analyzing and preventing pests, diseases, and other threats. Red palm weevils (RPWs) are a pest that causes much damage to CoTs and can cause devastation to vast areas of CoTs. CoT infestations by weevils are not visible to humans until they reach an advanced stage. To overcome this issue, a novel Internet of Things (IoT)-based COMICS model is proposed for detecting RPWs in CoT utilizing a microphone sensor and a stem image in the early stage. Environmental data are captured using an image capture device and a coconut microphone sensor. Initially, the input CoT images are denoised using the SavitzkyGolay (SavG) filter to reduce the noise artifacts. Then, the denoised CoT images are fed into the detection process using CAB-RetinaNet to detect damaged trees (DTs) and no DTs (NDTs). The coconut microphone sensor collects the audio signals, which are transformed into spectrograms. That signal is identified as the sound of the RPW, and the positional encoding self-attention bi-directional gated recurrent unit (PEN-BiGRU) model is used to determine the presence or absence of RPW. The outputs from both visual and audio analyses are fused in a decision-making module that uses fuzzy rules to evaluate CoT health and notify farmers through an IoT Blynk App. According to the collected dataset, the proposed COMICS model's performance analysis yields a total accuracy (ACY) of 99.54%. The proposed COMICS model increases the total ACY by 2.68%, 8.57%, 7.87%, 7.07%, and 38.34% for YOLOv5, ResNet50, YOLOv3, CocoNet model, and ResNet, respectively.



Index Terms—Coconut tree (CoT) disease, deep learning (DL), positional encoding self-attention bi-directional gated recurrent unit (PEN-BiGRU), RetinaNet, Savitzky–Golay (SavG) filter.

I. INTRODUCTION

COCONUT plants are vital to the agricultural economy in tropical regions, providing essential products such as coconuts, coconut oil, and copra [1]. Time-consuming and

frequently lacking in the accuracy needed for early detection, traditional illness diagnosis techniques rely mostly on visual inspections and manual interventions [2], [3], [4]. A major pest of palm trees, red palm weevil (RPW), *Rhynchophorus ferrugineus*, damages the tree significantly and costs farmers a lot of money [5], [6], [7].

The Internet of Things (IoT) [8] is revolutionizing modern agriculture by allowing real-time monitoring and data-driven decisions. A smart farming system based on IoT captures and analyzes data about environmental and agricultural conditions from sensors, drones, and automated systems. The development of precision agriculture technologies has enhanced irrigation, fertilization, and pest control [9], and reduced resource wastage while improving yield quality.

IoT helped to address these challenges by integrating advanced disease detection technologies [10], [11]. Sensors such as spectral cameras, temperature, humidity, and soil moisture were installed on the coconut plantations to gather data in real time [12]. These sensors can detect environmental conditions that may foster disease, enabling farmers to monitor plant conditions continuously [13]. Moreover, the gathered data were analyzed using machine learning (ML) [14] and

Received 18 July 2025; accepted 21 July 2025. Date of publication 24 November 2025; date of current version 14 January 2026. The associate editor coordinating the review of this article and approving it for publication was Prof. Liang-Bi Chen. (Corresponding author: L. K. Sowmya Sundari.)

L. K. Sowmya Sundari and B. Muthu Kumar are with the School of Computing and Information Technology, REVA University, Bengaluru, Karnataka 560064, India (e-mail: sowmyasundari.lk@reva.edu.in; muthu122@gmail.com).

Mallikarjun M. Kodabagi is with the Department of Computer Science and Engineering, CMR University, Bengaluru, Karnataka 562149, India (e-mail: mmkodabagi@gmail.com).

Mohammed Altaf Ahmed is with the Department of Computer Engineering, College of Computer Engineering and Sciences, Prince Sattam Bin Abdulaziz University (PSAU), Al-Kharj 16278, Saudi Arabia (e-mail: m.altaf@psau.edu.sa).

BalaAnand Muthu is with the Department of Computer Science and Engineering, Tagore Institute of Engineering and Technology, Salem, Tamil Nadu 636112, India (e-mail: Balavdy@gmail.com).

A. Ahilan is with the Department of Electronics and Communication Engineering, PSN College of Engineering and Technology, Tirunelveli, Tamil Nadu 627152, India (e-mail: listentoahil@gmail.com).

Digital Object Identifier 10.1109/JSEN.2025.3632878

deep learning (DL) methods in order to detect disease symptoms and diagnose possible outbreaks [15], [16], [17]. For training, coconut stem infections need a lot of annotated data, and getting that data was costly and time-consuming [18], [19]. The main contribution of the research is as follows.

- 1) In this research, a novel IoT-based COMICS model is proposed for detecting RPW in coconut trees (CoTs) using microphone sensor and stem image.
- 2) The denoised CoT images are fed into the detection process using CAB-RetinaNet to detect damaged tree (DT) and no DT (NDT).
- 3) That signal is identified as the sound of the RPW, and the positional encoding self-attention bi-directional gated recurrent unit (PEN-BiGRU) model is used to determine the presence or absence of RPW.
- 4) The outputs from both visual and audio analyses are fused in a decision-making module that uses fuzzy rules to evaluate CoT health and notify farmers through an IoT Blynk App.

The rest of this article is arranged as follows. The literature survey is summarized in Section II. The proposed methodology is illustrated in Section III. Section IV represents the experimental result and discussion. Section V represents the conclusion and future work.

II. LITERATURE SURVEY

In recent days, many techniques have been published for coconut stem disease detection based on IoT, DL, and ML approaches. Some of the relevant studies have been briefly discussed in this section.

Singh et al. [20] developed a DL and image processing to identify RPW pest infections, leaf blight diseases, and stem bleeding diseases in CoTs. The main purpose of the deep 2-D convolutional neural network (CNN) was to detect illnesses and pest infections. Wijethunga et al. [21] suggested three models for several facets of coconut farming: prediction of coconut plant development, ideal classification for coconut growth, and soil classification. The yellowing of leaves was detected using the YOLOv5 object detection method, which had an average precision of 96.87%.

Iqbal et al. [22] identified CoT in aerial images made available by the World Bank's AI competition, which was held in association with OpenAerialMap and WeRobotics. The Mask region-based CNN (R-CNN) method was utilized to identify and segment CoT. Utilizing architectures based on ResNet50 and ResNet101, the Mask R-CNN model was used to segment the data. In 2023, Roshan et al. [23] suggested to automatically recognize the disease afflicting CoTs from images of their leaves. The feature extraction stage involves retrieving pertinent attributes from the previously processed image using a deep CNN.

Yarak et al. [24] suggested a different method of managing oil palm trees by using Faster-RCNN for automatic detection and health classification in combination with high-resolution images. With ground sampling, ResNet-50 achieved $F1$ -scores of 97.67% (healthy) and 95.30% (unhealthy) using visual interpretation; the scores were 95.09%, 92.07%, and

86.96%. Samseemoung et al. [25] proposed a crane, chemical tanks and pumps, motors, motion and crane control systems, and nozzles with remote monitoring systems. The findings indicated ideal design parameters for CoTs that were 5–9 m tall. Karunarathna et al. [26] proposed a DL to identify the spatial distribution of individual CoTs in combination with uncrewed aerial vehicle (UAV) remote sensing. The outcomes indicate that the YOLOv3 model attained an average accuracy of 91.7%.

Jenifer and Natarajan [27] introduced a hybrid method that CocoNet uses to identify coconut farms. According to experimental findings, the suggested two-phase CocoNet model achieves 92.5% classification accuracy and 89.1% change detection accuracy. Megalingam et al. [28] suggested the Hyper Tuning SVM classification technique known as MIN-SVM, which is based on the Modified Inception Net. The MINSVM outperformed existing methods with an accuracy of 95.35%.

In 2024, Athira et al. [29] emphasize the use of high-resolution satellite imagery to evaluate CoT populations in Kanyakumari District, south India, both before and after the Ockhi cyclone. The CoT in the image is counted using a DL method based on an object identification framework (YOLO v8). Rasool et al. [30] suggested an identifying difficulty of RPW control in an early stage of infestation. The outcomes showed that the visual RPW detection method had the accuracy of 87%, the radar 2000 of 77%, and the radar 900 of 73%.

Alsanea et al. [31] presented a DL model that classifies and differentiates the RPW from other insects that live in palm tree environments. To find the RPW, an R-CNN method was employed, which creates bounding boxes around the image. Na et al. [32] developed pheromones, food baits, and environmentally acceptable catching methods, and researchers studied the RPW habitat in the United Arab Emirates. Its 2.69 ± 0.07 times greater capture rates than the commercially available pheromone validated the newly synthesized pheromone's potential as an RPW.

Arasi et al. [33] describe the Bird Swarm Algorithm with DL as a sophisticated RPW detection system. For the hyperparameter tuning procedure, the IRPWD-BSADL model uses BSA to improve the recognition outcomes. Albraikan et al. [34] created the Gorilla Troops Optimizer with a DL approach for automatic RPW detection. Additionally, the RPWD-GTODL technique's detection performance was enhanced by the hyperparameter design of the GTO method for the MobileNetv2 model.

Kagan et al. [35] suggested monitoring palm trees affected by RPW using advanced DL algorithms and data from aerial and street-level images. This approach has the potential to completely transform the outdated methods currently used to manage RPWs by offering far more affordable, effective, and efficient monitoring. Montero et al. [36] created a model that uses computer vision and the DL model to identify yellowing, flaccidity, drying leaflets, caterpillars, and leaflets in CoTs. In order to enhance agricultural disease control initiatives, the suggested model is a dependable diagnostic for efficiently identifying illnesses and infestations in coconut.

The FNN–SWO model proposed by El-Shafeiy et al. [37] evaluated using an adaptive network and fuzzy rules.

The three data subsets have been used to compare the suggested FNN–SWO model to a number of the most widely used traditional methods. Barman et al. [38] identify insect damage accurately through the use of color and texture analysis. The gray-level run length matrix (GLRLM) and gray-level co-occurrence matrix (GLCM) techniques are used to extract the textural features of the damages. Manoharan et al. [39] created the YOLOv9 model to identify deficits in coconut plants' macro and micronutrients. The severity index (SI) of these defects is also determined using the SI calculation model (SICM), which is introduced in conjunction with these abilities. The accuracy, precision, and recall of YOLOv9 were 80%, 98.59%, and 80.37%, respectively, better than those of other models.

Existing studies focus on either DL or IoT, while the proposed method combines both IoT and DL for simultaneous detection of coconut stem diseases and RPW that provide efficient results with better accuracy. The main goal of the research is to develop a novel IoT-based COMICS model for detecting RPW in CoTs using a microphone sensor and stem image.

III. PROPOSED METHODOLOGY

In this research, a novel IoT-based COMICS model is proposed for detecting RPW in CoTs using a microphone sensor and stem image [see Fig. 1(a)].

A. INMP441 Sensor

The INMP441 sensor is a high-performance low-noise digital microphone used for detecting the sound of RPW. An RPW detection system captures acoustic signals with high sensitivity and precision, making it ideal for identifying their unique acoustic patterns. The sensor is mounted on a custom-designed printed circuit board (PCB) for easy integration with IoT systems. Its digital output ensures minimal signal distortion, enabling accurate sound analysis and reliable pest detection in real-time monitoring applications for CoT health management. In situ installation images are provided to demonstrate the practical setup of the proposed COMICS model in a coconut plantation. The camera is mounted at a 45° angle and positioned at a height of 1.5 m to capture the stem image clearly. The microphone sensor insertion point is embedded approximately 30 cm into the CoT stem at a height of 1.2 m. The setup also includes a solar panel and battery system installed on a nearby pole, supplying uninterrupted power to the sensors and IoT unit. The labeled images provide an overview of the system's deployment ability in agricultural settings. The INMP441 sensor is used in the digital microphone sensor shown in Fig. 1(b).

B. Dataset Description

In this work, the CoT (Cocos nucifera) Disease Dataset [40] was used to detect the CoT disease. The CoT Disease Dataset provides images that can be used in DL applications to detect and classify coconut disease. In the “CoT Disease Dataset,”

there are high-resolution images with a width and a height of 768 and 1024 pixels, respectively. In order for the CoT diseases to be clearly seen, they are displayed at a resolution of 72 dpi. The dataset includes 5798 images across five disease classes: Bud Rot, Leaf Rot, Bud Root Dropping, Stem Bleeding, and Gray Leaf Spot. This diversity is necessary to train and evaluate DL algorithms capable of detecting and classifying CoT diseases. The proposed COMICS model was assessed with the cocos nucifera disease dataset, which is separated into 20% for testing and 80% for training.

C. SavG Filter for Preprocessing

In this section, the input CoT images are denoised using a SavG filter to reduce the noise artifacts. By applying this filter, the coconut images are enhanced allowing for improved clarity and better representation of the structural details of the CoTs. A collection of m convolution coefficients C_k is used to represent the image, which consists of a set of n intensity values $Y = \{Y_1 \dots Y_n\}$. The Savitzky–Golay (SavG) filter approximates each point Y_i by a weighted sum of points using the following convolution:

$$Y_i = \sum_{k=-(m-1)/2}^{(m-1)/2} C_k \cdot y_{i+k} \text{ for } \frac{m-1}{2} \leq i \leq n - \frac{m-1}{2}. \quad (1)$$

To evaluate the similarity between the original image $\{X_1 \dots X_n\}$ and the filtered image $Y = \{Y_1 \dots Y_n\}$, the correlation coefficient (COR) is computed as follows:

$$\text{COR} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \cdot \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}} \quad (2)$$

where datasets $\bar{X}$ and $\bar{Y}$ are the sample means of the datasets X and Y , respectively. When the frame size m at its maximum and the synthetic and SavG-filtered images have a 1.00 correlation, the filter is considered optimally tuned.

D. CoT Damage Detection Using CAB-RetinaNet

The preprocessed CoT images are fed into the detection process using CAB-RetinaNet to detect DT and NDT. RetinaNet struggles with detecting trees of varying sizes, especially small or occluded trees, due to its fixed-scale feature maps. Compared to lightweight models, RetinaNet requires significant computational resources, making real-time tree damage assessment challenging in field applications. Variations in lighting shadows and seasonal changes can affect RetinaNet's ability to distinguish between damaged and non-DTs accurately. The CAB-RetinaNet refines feature representation, resulting in fewer false positives and negatives, making it more accurate at identifying tree damage. A feature pyramid network (FPN) is used to collect features from the C3–C5 layers and is converted into five useful layers: P3–P7. The main feature extraction network employed ResNeXt, which contains fewer parameters. Through a 1×1 convolutional layer, the outputs from every group are combined. Cardinality refers to the magnitude of the set of transformations.

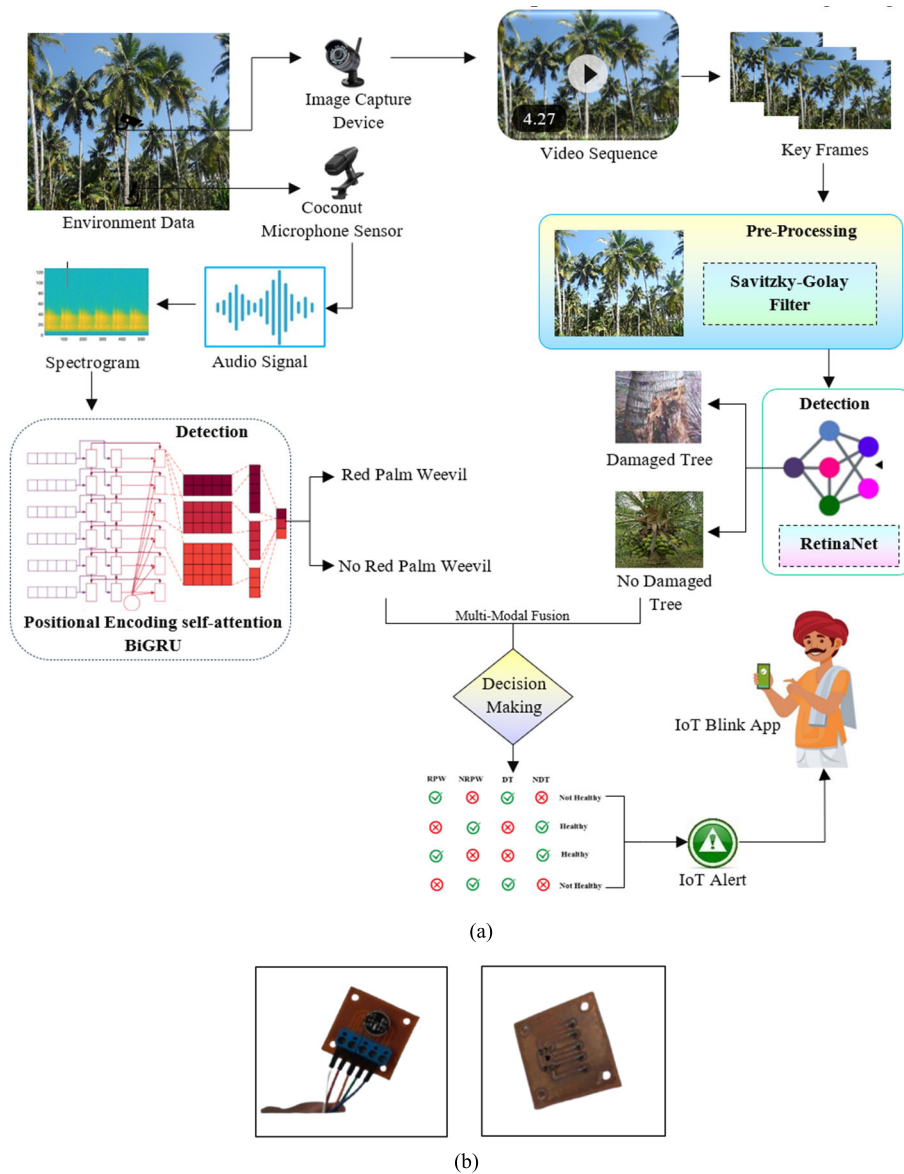


Fig. 1. (a) Proposed IoT-based COMICS model. (b) Digital microphone sensor using INMP441 sensor.

1) **CAB-RetinaNet**: RetinaNet outperforms classical networks but struggles with accurately detecting coconut stem diseases in images. The attention module with a B-module is an enhanced multiscale feature fusion channel in CAB-RetinaNet. ResNeXt extracts multiscale features from the C3 and C4 layers to address small-target gradient problems. ResNeXt is used to extract the final feature, the C5 layer. The attention module is then added by CAB-RetinaNet to create enhanced features with rich spatial information. Fig. 2 displays the overall CAB-RetinaNet architecture.

E. RPW Sound Detection Using PEN-BiGRU Model

The input audio signal is identified as the sound of the RPW, and the PEN-BiGRU model is used to determine the presence or absence of RPW. BiGRU requires more memory and processing power than traditional models, making real-time detection difficult on low-power devices. RPW sounds and environmental noise were hard to distinguish for BiGRU,

leading to false positives and lowered accuracy. PEN-BiGRU integrates positional encoding to differentiate RPW sounds from environmental noise. PEN-BiGRU prioritizes important features by increasing the efficiency of RPW sound detection.

1) **Positional Encoding**: The PEN-BiGRU model is used for detecting RPW in signals obtained from coconut monitoring sensors in CoTs.

2) **BiGRU Layer**: A gated recurrent unit network (GRU) is a specialized form of recurrent neural network (RNN) designed to address the limitations of traditional RNNs. GRU is a simple LSTM neural network that was efficiently determined while preserving the effectiveness of LSTM neural networks. Here, at each time step t , the model receives an input vector x_t and the previous hidden state h_{t-1} . The reset gate r_t , update gate z_t , and $\tilde{h}_t$ are calculated by applying the tanh activation to the reset-modulated earlier hidden state and the current input. The candidate state weighted by the update gate is combined with

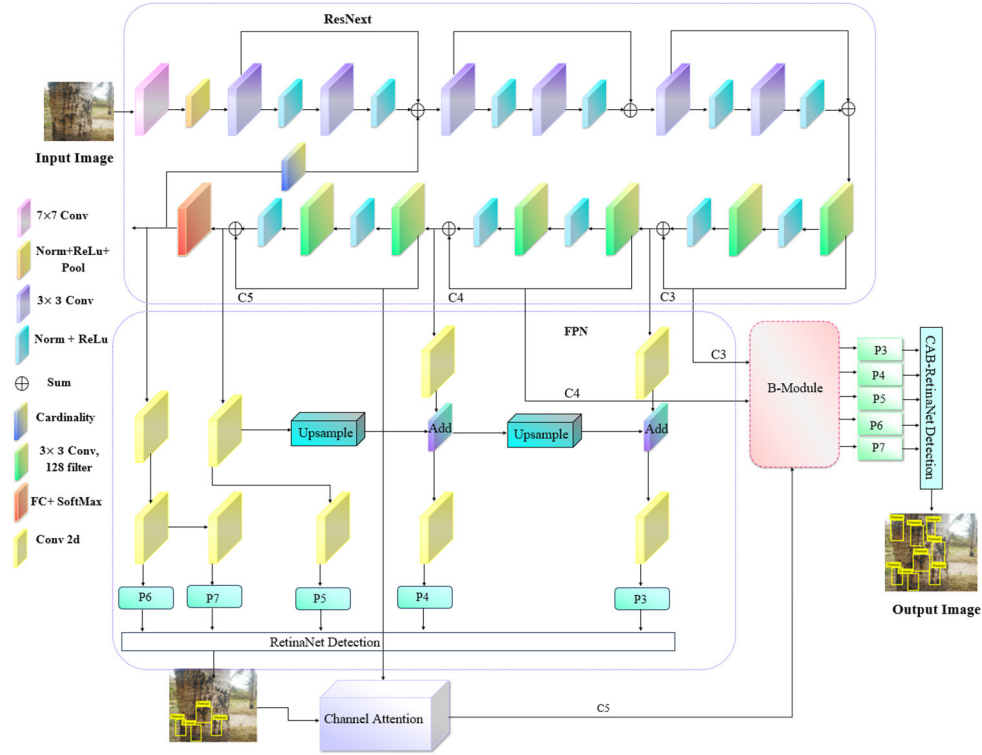


Fig. 2. Architecture of CAB-RetinaNet.

the previous hidden state h_t

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (3)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (4)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (5)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (6)$$

where W_z , W_r , and W_h , and U_z , U_r , and U_h are associated with the input and the hidden state, and b_z , b_r , and b_h are the corresponding bias vectors. The operator $\odot$ denotes the element-wise multiplication.

The BiGRU structure consists of two hidden layers: forward and backward. Following a symmetrical combination with both of those state vectors as shown next:

$$h_t = \text{concat}(\vec{h}_t, \overleftarrow{h}_t) \quad (7)$$

where $\vec{h}_t$ is the forward GRU and $\overleftarrow{h}_t$ denotes the backward GRU that receives each data pattern as input, generating two symmetric hidden layer state vectors.

3) Self-Attention Layer: The attention function maps a query (Q) to a set of key-value pairs. The self-attention output is

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (8)$$

where QK^T denotes the similarity between each query and all keys, $(d_k)^{1/2}$ scales the dot product to prevent large values, Softmax converts similarities into attention weights, and V denotes attention weights to the value vectors.

Fig. 3 shows the flowchart of the proposed COMICS model, comprising three core phases: installation, calibration, and

cloud connection. In the installation phase, the INMP441 digital microphone sensor is embedded into the coconut stem, a camera is mounted at a fixed angle, and a solar-powered system is set up to ensure an uninterrupted energy supply. During the calibration phase, trial recordings are performed to distinguish RPW sounds from background noise, camera settings are optimized, and the CAB-RetinaNet and PEN-BiGRU models are validated for detecting tree damage and weevil activity. Finally, in the cloud connection phase, the microcontroller is linked via GSM, enabling real-time data transmission to the Blynk IoT app, where alerts and health reports are delivered to farmers using fuzzy decision logic. The model's potential for practical use in precision agriculture is validated by this sequential process.

F. Fuzzy-Based Decision-Making

The outputs from both visual and audio analyses are fused in a decision-making module that uses fuzzy rules to evaluate CoT health and notify farmers through an IoT Blynk App. Inputs between 0 and 1 are accepted in this Mamdani inference system with Gaussian membership functions (MFs). A fuzzification unit converts the four detection outputs (DT, NDT, RPW, and NRPW) into fuzzy measures

$$\text{Fuzzy} = \begin{cases} 1, & \text{if } (K_a, K_b) \geq (K'_a, K'_b) \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

where K_a and K_b are the things found in the CoT, K'_a and K'_b are the tree detections, and fuzzy is defined as a fuzzy decision-making variable.

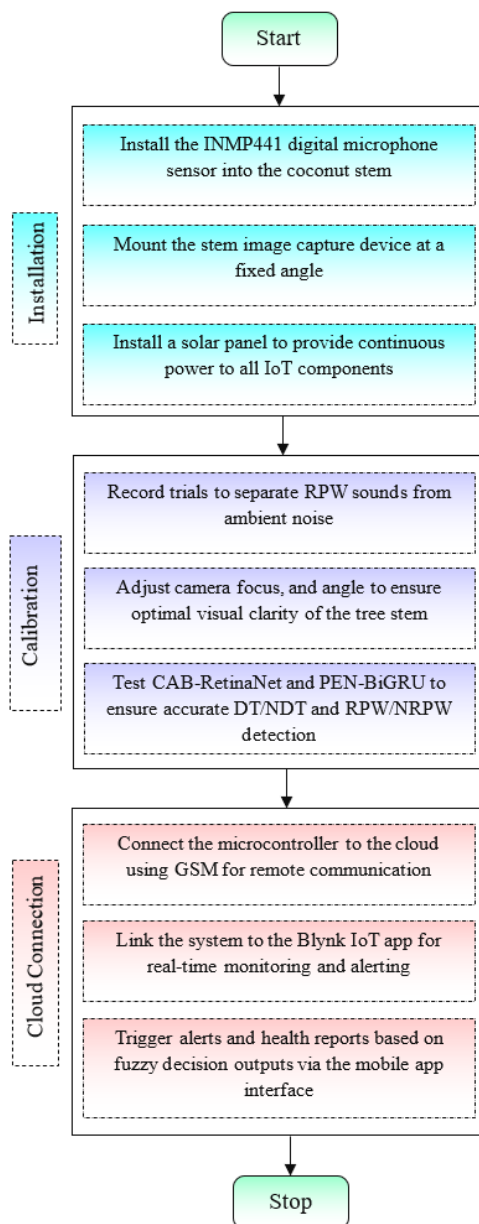


Fig. 3. Flowchart of the proposed COMICS model.

TABLE I
RULES FOR FUZZY INFERENCE SYSTEM

Rules	RPW	NRPW	DT	NDT	Decision	Output
1	✓	X	✓	X	Not Healthy	Alert
2	✓	X	X	✓	Healthy	No Alert
3	X	✓	✓	X	Not Healthy	Alert
4	X	✓	X	✓	Healthy	No Alert

When any disease in the CoT health notifies farmers through an IoT Blynk App, in Table I, the fuzzy rules are separated into four classes: healthy, unhealthy, RPW, and NRPW. The detection of DT and RPWs is dependent on four fuzzy rules.

The results of the CAB-Retina network are regulated by a fuzzy interference regulation. PEN-BiGRU and CAB-Retina network outcomes are adjusted using fuzzy interference regulation. CoTs are protected from infestation by integrating the

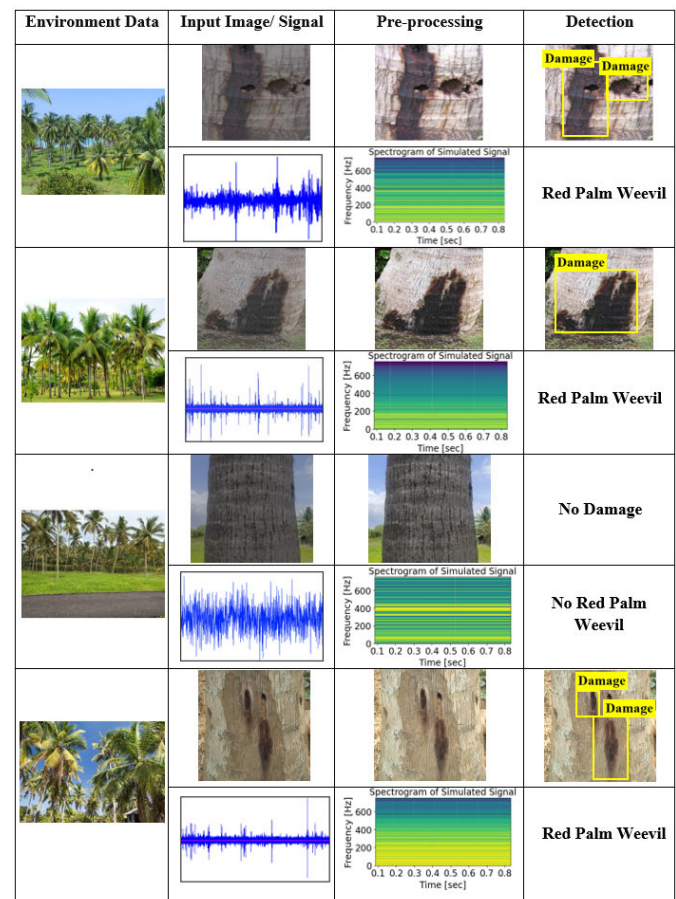


Fig. 4. Experimental outcomes of the proposed COMICS model.

two MFs into the fuzzy system and notifying farmers via an IoT Blynk App.

IV. RESULTS AND DISCUSSION

The experimental study was implemented using Spyder, an Anaconda Navigator running under Windows 10 OS with an Intel i3 processor running at 2.10 GHz and 8-GB RAM.

Fig. 4 illustrates the experimental results of the proposed COMICS model for detecting damaged CoTs. Column 1 shows the environmental data of CoT plantations representing various environmental contexts. The second column shows the input CoT images for visual analysis and RPW sound signals captured using a coconut microphone sensor from the CoTs. The third column illustrates the preprocessing stage, where images are enhanced using the SavG filter and acoustic signals are converted into spectrograms to extract time–frequency features relevant to RPW. Finally, the fourth column demonstrates the detection of damaged is localized using bounding boxes, while spectrogram-based analysis classifies the presence or absence of RPW. The fused visual and acoustic signal supports accurate and early identification of RPW infestation in CoTs.

A farmer alert system is developed to access real-time data acquisition from the CoT health notify and send alert to the farmer, if any abnormal situation is raised. The applied fuzzy rule makes a recommendation to the farmer to check the status

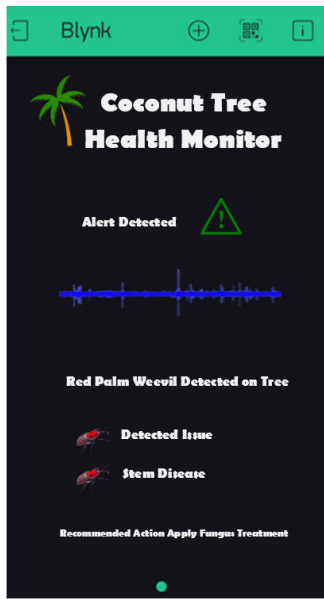


Fig. 5. Real-time experimental setup for RPW detection.

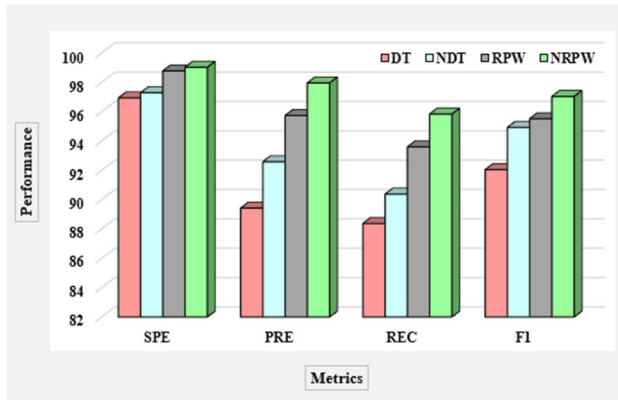


Fig. 6. Graphical evaluation of the proposed COMICS model.

of the coconut field. Fig. 5 shows the real-time experimental setup of the proposed model.

A. Performance Analysis

The proposed COMICS model was evaluated in an experimental study using the *cocos nucifera* illness dataset, which is divided into 20% for testing and 80% for training. The COMICS model evaluates $F1$ -score ($F1$), recall (REC), specificity (SPE), accuracy (ACC), and precision (PRE). Fig. 6 shows the graphical evaluation of the proposed COMICS model. The proposed COMICS model's overall ACC, as determined by performance analysis using the collected dataset, is 99.54%.

The accuracy and epochs are positioned on opposite axes in Fig. 7(a), which represents the ACC curve. As the number of epochs rises, so does the model's ACC. The model's loss is represented by the epoch versus loss curve in Fig. 7(b).

Fig. 8(a) represents the time-domain and spectrogram analysis of the RPW sound signal over three different 8-h periods. Fig. 8(b) represents the spectrograms of the RPW signal, highlighting its frequency components over time. The

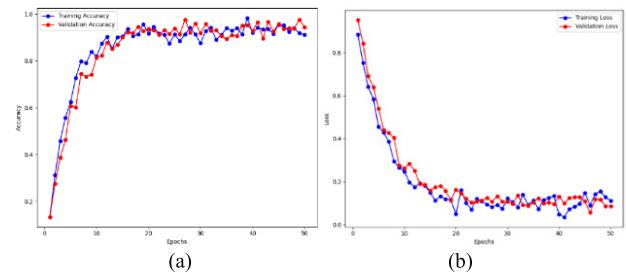


Fig. 7. (a) Accuracy curve and (b) loss curve of the proposed COMICS model.

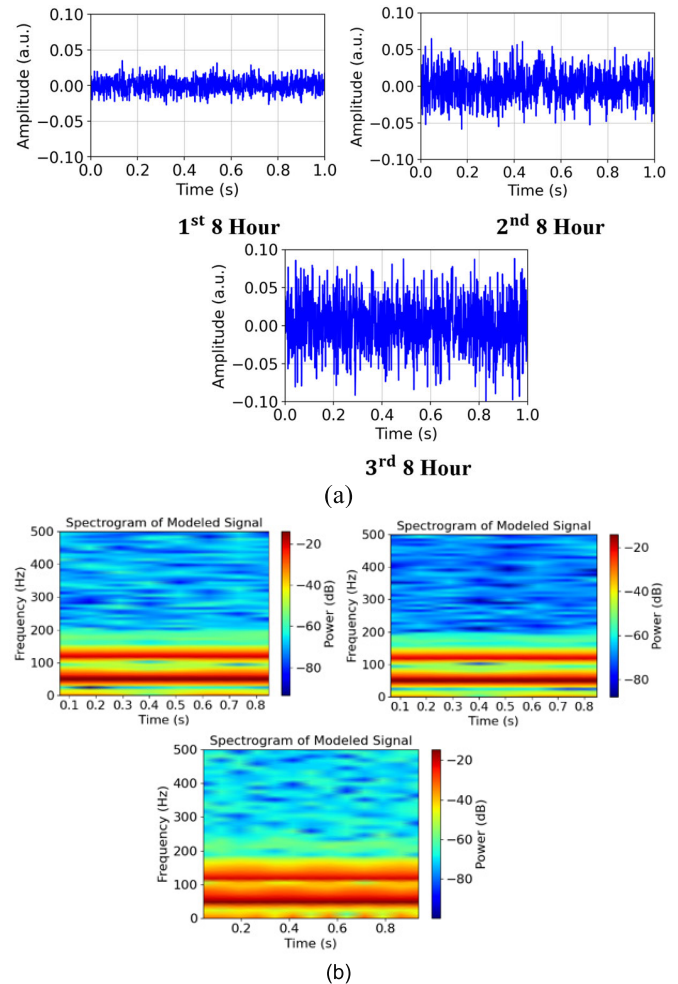


Fig. 8. Signal analysis over three 8-h intervals. (a) Time-domain RPW signals. (b) Corresponding spectrograms.

spectrograms show a high degree of consistency at low frequencies, specifically 100–200 Hz for typical RPW activity. Additionally, higher frequency components observed in later periods indicate noise or overlapping signals by emphasizing the importance of time–frequency analysis. As the power spectrum density changes over time, the energy density increases in the later segments, which was indicative of the behavioral or environmental changes. The spectrograms of the RPW sounds effectively reproduce the dynamic characteristics of the modeling approach.

Table II presents a comparative analysis of various CoT disease datasets based on three performance metrics like

TABLE II
PERFORMANCE ANALYSIS OF THE PROPOSED MODEL
BASED ON DIFFERENT DATASETS

Datasets	ACC	SPE	F1
CocoPalm Dataset	88.39%	87.43%	86.94%
CoconutNet Dataset	90.28%	89.58%	89.48%
Tropical Tree Disease Dataset	92.41%	92.34%	90.24%
Coconut Disease Detection Dataset	95.73%	94.82%	92.59%
Coconut Tree Disease Dataset (Ours)	99.54%	98.00%	94.87%

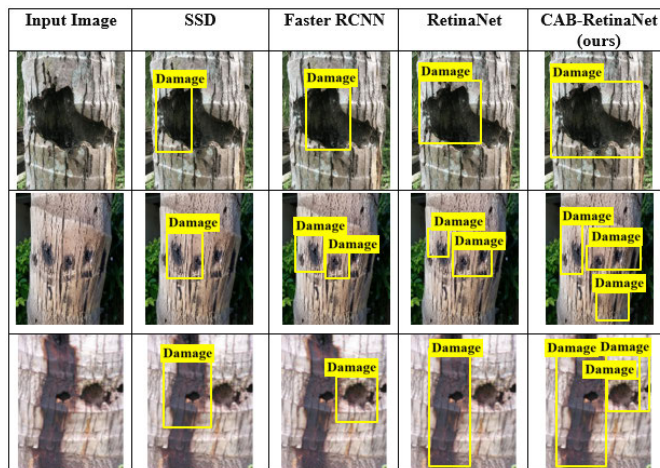


Fig. 9. Comparison of the existing framework with the proposed framework for coconut stem disease detection.

ACC, SPE, and $F1$. The findings show that the proposed dataset performs better than current datasets on every metric, with the greatest $F1$ -score (94.87%), accuracy (99.54%), and specificity (98.00%). Based on the CoT disease dataset, the COMICS model performs improved than the current methods.

B. Comparative Analysis

A comparative analysis was conducted to compare the proposed COMICS model to various existing techniques.

Fig. 9 illustrates the comparison output of the existing models and the proposed model for defect detection. The input CoT images from each model are displayed in each row. The subsequent columns display the detection results of each model, along with damage regions. SSD and Faster R-CNN identify the primary damage areas, but they often fail to locate small or multiple damaged areas accurately. The existing retinal network captures multiple damage spots, but some areas remain incorrectly identified. The proposed CAB-RetinaNet effectively detects all damage regions and localizes each one in multiple instances at the same time. In comparison to the object detection approach, the ACC rate demonstrates the superior efficacy of the proposed approach for defect detection.

Fig. 10 shows the assessment of different DL techniques. The performance of the COMICS model in severe weather scenarios, including heavy rainfall or snowfall, does not achieve good results. The computational efficiency and real-time feasibility of the model for large-scale agricultural deployment require further evaluation.

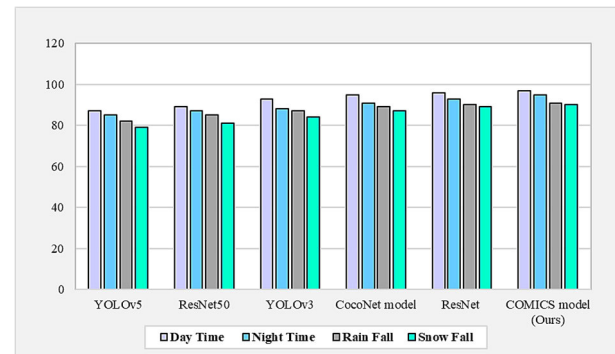


Fig. 10. Accuracy comparison of object detection models under various environmental conditions.

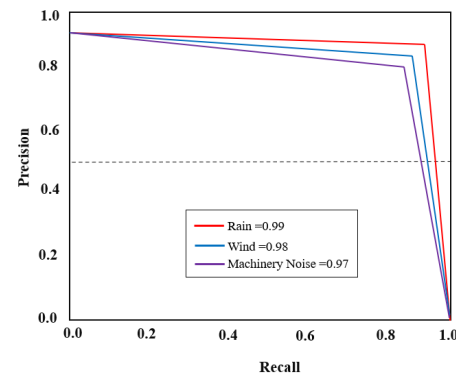


Fig. 11. ROC curve for the proposed model based on different scenarios.

Fig. 11 shows the precision–recall curves under different noise conditions, indicating stable performance. The experimental results under rain, strong wind, and farm machinery noise scenarios indicate that the proposed model maintains high precision and recall across all conditions. However, slight performance degradation is observed in the presence of machinery noise.

Fig. 12 displays the confusion matrix for the COMICS model based on various climatic conditions. The confusion matrix reveals that most misclassifications occur between acoustically similar noise types, particularly between wind and machinery sounds. Overall, the model demonstrates strong robustness against environmental noise, though future enhancements could focus on improving discrimination in highly overlapping acoustic scenarios.

Table III shows the average current draw and estimated battery life of hardware modules in the proposed COMICS model. The INMP441 microphone operates for up to 122 days on a 4000-mAh battery due to its short duty cycle and low current consumption. Meanwhile, the ESP32 MCU with audio processing consumes more power during operation, reducing its battery life to approximately 5.5 days. The significance of incorporating solar-powered power systems or optimizing duty cycles to increase operating duration and decrease maintenance intervals in field deployments is confirmed by these findings.

Table IV displays the evaluation of the traditional object detection networks. The proposed CAB-RetinaNet achieved the ACC was improved by 7.21%, 6.46%, 4.54%, and

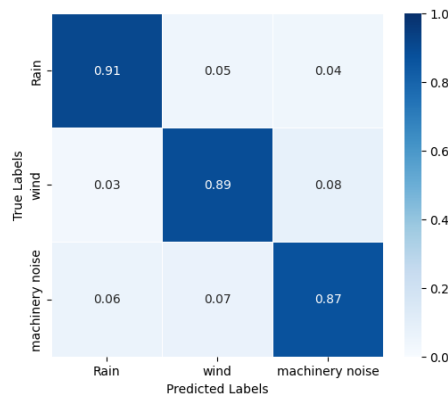


Fig. 12. Confusion matrix for the proposed model based on different climatic conditions.

TABLE III

AVERAGE CURRENT DRAW AND BATTERY LIFE ESTIMATION

Hardware Module	Idle Current (mA)	Active Current (mA)	Duty Cycle	Average Current (mA)	Estimated Battery Life (4,000 mAh)
INMP441 Microphone	0.8	3.2	20%	1.36	122 days
ESP32 MCU + Audio Processing	15	160	10%	30.5	5.5 days
Camera Module	5	120	5%	10.75	15.5 days
GSM / Wi-Fi Module	10	180	5%	19.5	8.5 days
Total System	—	—	—	62.11 mA	2.7 days

TABLE IV

COMPARISON OF THE EXISTING AND PROPOSED MODELS

Networks	ACC	SPE	PRE	REC	F1
SSD	92.36	90.37	88.27	79.83	87.84
Faster RCNN	93.10	91.93	89.02	81.27	89.27
Mask RCNN	95.02	93.18	90.28	84.20	91.73
RetinaNet	97.69	95.01	91.48	89.98	92.14
CAB-RetinaNet (Ours)	99.54	98.00	93.92	92.04	94.87

1.85% over SSD, Faster RCNN, Mask RCNN, and RetinaNet, respectively. Based on the previously specified network metrics, CAB-Retina performed better than the other detection networks.

The comparison between the proposed COMICS model and several existing models is displayed in Table V. The proposed COMICS model outperformed all others with an ACC of 99.54% and a p-value of 0.008, indicating both strong performance and statistical significance. The proposed COMICS model increases the overall accuracy by 2.68%, 8.57%, 7.87%, 7.07%, and 38.34% for YOLOv5, ResNet50, YOLOv3, CocoNet model, and ResNet, respectively. The proposed COMICS model detects CoT diseases with greater ACC and statistical significance, according to this overall comparative analysis.

TABLE V

COMPARISON OF THE PROPOSED COMICS MODEL VERSUS THE EXISTING MODEL

Author	Techniques	Accuracy	p-value
Wijethunga, C.D., et al., [21]	YOLOv5	96.87%	0.029
Iqbal, M.S., et al., [22]	ResNet50	91%	0.015
Karunaratna, et al., [26]	YOLOv3	91.7%	0.048
Jenifer, A.E. [27]	CocoNet model	92.5%	0.087
Megalingam, R.K., [28]	ResNet	71.95%	0.095
Proposed	COMICS model	99.54%	0.008

V. CONCLUSION

This article proposed a novel IoT-based COMICS model for detecting RPW in CoTs utilizing a microphone sensor and stem image. Environmental data are captured using an image capture device and a coconut microphone sensor. The outputs from both visual and audio analyses are fused in a decision-making module that uses fuzzy rules to evaluate CoT health and notify farmers through an IoT Blynk App. The proposed COMICS model increases the overall ACC by 2.68%, 8.57%, 7.87%, 7.07%, and 38.34% for YOLOv5, ResNet50, YOLOv3, CocoNet model, and ResNet, respectively. Nevertheless, performance dropped slightly when different noise scenarios overlapped, such as wind and machinery noises. Future research will explore adaptive multiband thresholding and context-aware noise profiling to enable dynamic sensitivity adjustments based on environmental changes. Additionally, wavelet-based filters and attention-guided denoising modules will be included to improve robustness and reduce false positives.

REFERENCES

- [1] E. Honlah, A. Yao Segbefia, D. Forkuor, and K. Abass, "Trading future sustainable livelihoods for present gain: Labour dynamics, opportunities, challenges and sustainability of virgin coconut oil processing businesses in the wake of coconut exports from southwestern Ghana," *Cogent Food Agricult.*, vol. 10, no. 1, Dec. 2024, Art. no. 2395476.
- [2] F. Vieira et al., "Coconut waste: Discovering sustainable approaches to advance a circular economy," *Sustainability*, vol. 16, no. 7, p. 3066, Apr. 2024.
- [3] S. Bharathi and P. Harini, "Early detection of diseases in coconut tree leaves," in *Proc. 6th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Mar. 2020, pp. 1265–1268.
- [4] M. Tzec-Simá et al., "Potential of omics to control diseases and pests in the coconut tree," *Agronomy*, vol. 12, no. 12, p. 3164, Dec. 2022.
- [5] M. A. Rahman, M. N. Alam, M. N. A. Siddiquee, M. E. Reza, and M. S. Islam, "The prevalence and control of Red Palm Weevil, *Rhynchophorus ferrugineus* oliver (coleoptera: Curculionidae) infestations on date palms in Bangladesh," *Int. J.*, vol. 3, no. 2, pp. 22–28, 2023.
- [6] R. Raja Kumar, J. Athimoolam, A. Appathurai, and S. Rajendiran, "Novel segmentation and classification algorithm for detection of tomato leaf disease," *Concurrency Comput., Pract. Exper.*, vol. 35, no. 12, p. e7674, May 2023.
- [7] W. Bouhila, A. Alzahem, A. Koubaa, B. Benjdira, and A. Ammar, "Early detection of Red Palm Weevil infestations using deep learning classification of acoustic signals," *Comput. Electron. Agricult.*, vol. 212, Sep. 2023, Art. no. 108154.

- [8] M. M. Atia, A. I. Gouda, M. G. Ismail, M. A.-M. Salem, and M. A. A. ElGhany, "Acoustic device for detecting Red Palm Weevil using deep learning and IoT," in *Proc. Int. Conf. Microelectron. (ICM)*, Dec. 2023, pp. 1–5.
- [9] A. Koubaa et al., "Smart palm: An IoT framework for Red Palm Weevil early detection," *Agronomy*, vol. 10, no. 7, p. 987, Jul. 2020.
- [10] B. B. Sinha and R. Dhanalakshmi, "Recent advancements and challenges of Internet of Things in smart agriculture: A survey," *Future Gener. Comput. Syst.*, vol. 126, pp. 169–184, Jan. 2022.
- [11] K. Sharma and S. K. Shivandu, "Integrating artificial intelligence and Internet of Things (IoT) for enhanced crop monitoring and management in precision agriculture," *Sensors Int.*, vol. 5, Jan. 2024, Art. no. 100292.
- [12] Y. Hajjaji, A. Alzahem, W. Boulila, I. R. Farah, and A. Koubaa, "Sustainable palm tree farming: Leveraging IoT and multi-modal data for early detection and mapping of Red Palm Weevil," *Proc. Comput. Sci.*, vol. 225, pp. 4952–4962, Jan. 2023.
- [13] N. Chamara, M. D. Islam, G. F. Bai, Y. Shi, and Y. Ge, "Ag-IoT for crop and environment monitoring: Past, present, and future," *Agric. Syst.*, vol. 203, Dec. 2022, Art. no. 103497.
- [14] A. G. Karegowda, D. Vaishnavi, and Y. Harshalatha, "Coconut disease detection using traditional ML classifiers with GLCM and SIFT features," in *Proc. 8th Int. Conf. Comput. Syst. Inf. Technol. Sustain. Solutions (CSITSS)*, Nov. 2024, pp. 1–6.
- [15] N. B. Shankara and P. Peddi, "Implementing efficient machine learning techniques to identify coconut leaves enhances the diagnostic accuracy of diseases," *Int. J. Res. Eng. Appl. Sci.*, vol. 9, no. 7, pp. 1–8, Jul. 2024.
- [16] S. Govindapillai and R. Ayyapazham, "Segmentation and classification of paddy leaf disease via deep learning networks," *Int. J. Data Sci. Artif. Intell.*, vol. 2, no. 5, pp. 161–169, 2024.
- [17] S. K. Brar, R. Sharma, S. Vats, and V. Kukreja, "A smart approach to coconut leaf spot disease classification using computer vision and deep learning technique," in *Proc. World Conf. Commun. Comput. (WCONF)*, Jul. 2023, pp. 1–6.
- [18] S. Savaş, "Application of deep ensemble learning for palm disease detection in smart agriculture," *Heliyon*, vol. 10, no. 17, Sep. 2024, Art. no. e37141.
- [19] M. Ayaz, M. Ammad-Uddin, Z. Sharif, A. Mansour, and E. M. Aggoune, "Internet-of-Things (IoT)-based smart agriculture: Toward making the fields talk," *IEEE Access*, vol. 7, pp. 129551–129583, 2019.
- [20] P. Singh, A. Verma, and J. S. R. Alex, "Disease and pest infection detection in coconut tree through deep learning techniques," *Comput. Electron. Agric.*, vol. 182, Mar. 2021, Art. no. 105986.
- [21] C. D. Wijethunga, K. C. Ishanka, S. D. N. Parindya, T. J. N. Priyadarshani, B. Harshanath, and S. Rajapaksha, "Coconut plant disease identified and management for agriculture crops using machine learning," *Int. J. Eng. Manage. Res.*, vol. 13, no. 5, pp. 79–88, Oct. 2023.
- [22] M. S. Iqbal, H. Ali, S. N. Tran, and T. Iqbal, "Coconut trees detection and segmentation in aerial imagery using mask region-based convolution neural network," *IET Comput. Vis.*, vol. 15, no. 6, pp. 428–439, Sep. 2021.
- [23] R. Roshan, S. Santhi, and R. Rishiram, "Design and development of coconut tree disease identification using deep learning techniques," Tech. Rep.
- [24] K. Yarak, A. Witayangkurn, K. Kritiyutanont, C. Arunplod, and R. Shibasaki, "Oil palm tree detection and health classification on high-resolution imagery using deep learning," *Agriculture*, vol. 11, no. 2, p. 183, Feb. 2021.
- [25] G. Samseemoung, P. Soni, and P. Suwan, "Development of a variable rate chemical sprayer for monitoring diseases and pests infestation in coconut plantations," *Agriculture*, vol. 7, no. 10, p. 89, Oct. 2017.
- [26] R. M. D. M. Karunarathna, K. H. N. Kulapathi, and K. L. P. I. Liyanage, "A case study on detecting and mapping individual coconut trees using YOLOv3 in conjunction with UAV," *Remote Sens. Smart Plantation Manag.*, 2022.
- [27] A. E. Jenifer and S. Natarajan, "CocoNet: A hybrid machine learning framework for coconut farm identification and its cyclonic damage assessment on bitemporal SAR images," *J. Appl. Remote Sens.*, vol. 15, no. 4, Jul. 2021, Art. no. 042408.
- [28] R. K. Megalingam, S. K. Manoharan, D. H. T. A. Babu, G. Sriram, K. Lokesh, and S. K. Sudheesh, "Coconut trees classification based on height, inclination, and orientation using MIN-SVM algorithm," *Neural Comput. Appl.*, vol. 35, no. 16, pp. 12055–12071, Jun. 2023.
- [29] A. R. Athira, K. G. Lakshmi, C. H. Genitha, and A. Tamizhselvi, "Deep learning based approach to assess the impact of cyclone Ockhi on coconut tree population using satellite images," in *Proc. Asian Conf. Remote*, 2024, pp. 1–9.
- [30] K. G. Rasool et al., "Evaluation of some non-invasive approaches for the detection of Red Palm Weevil infestation," *Saudi J. Biol. Sci.*, vol. 27, no. 1, pp. 401–406, 2020.
- [31] M. Alsanea, S. Habib, N. F. Khan, M. F. Alsharekh, M. Islam, and S. Khan, "A deep-learning model for real-time Red Palm Weevil detection and localization," *J. Imag.*, vol. 8, no. 6, p. 170, Jun. 2022.
- [32] S.-M. Na, G.-I. Im, W.-S. Lee, and D.-G. Kim, "Assessment of attractant combinations for the management of Red Palm Weevils (*Rhynchophorus ferrugineus*) in the United Arab Emirates," *Insects*, vol. 15, no. 4, p. 218, Mar. 2024.
- [33] M. A. Arasi, L. Almuqren, I. Issaoui, N. S. Almalki, A. Mahmud, and M. Assiri, "Enhancing Red Palm Weevil detection using bird swarm algorithm with deep learning model," *IEEE Access*, vol. 12, pp. 1542–1551, 2024.
- [34] A. A. Albraikan et al., "Automated Red Palm Weevil detection using gorilla troops optimizer with deep learning model," *IEEE Access*, vol. 11, pp. 71616–71623, 2023.
- [35] D. Kagan, G. Fuhrmann Alpert, and M. Fire, "Automatic large scale detection of Red Palm Weevil infestation using street view images," *ISPRS J. Photogramm. Remote Sens.*, vol. 182, pp. 122–133, Dec. 2021.
- [36] C. D. Montero, B. J. Gundul, J. Frayco, and J. Candia Jr., "Convolutional neural network-based image classification for improved coconut disease identification," *SDSSU Multidisciplinary Res. J.*, vol. 11, no. 1, pp. 17–21, Oct. 2023.
- [37] E. El-Shafeiy, A. A. Abohany, W. M. Elmessery, and A. A. A. El-Mageed, "Estimation of coconut maturity based on fuzzy neural network and sperm whale optimization," *Neural Comput. Appl.*, vol. 35, no. 26, pp. 19541–19564, Sep. 2023.
- [38] U. Barman, C. Pathak, and N. K. Mazumder, "Comparative assessment of pest damage identification of coconut plant using damage texture and color analysis," *Multimedia Tools Appl.*, vol. 82, no. 16, pp. 25083–25105, Jul. 2023.
- [39] S. K. Manoharan, R. K. Megalingam, A. Gopika, G. Jogesh, K. Aryan, and A. R. Kunnambath, "AI based early identification and severity detection of nutrient deficiencies in coconut trees," *Smart Agric. Technol.*, vol. 9, Dec. 2024, Art. no. 100575.
- [40] S. Thite, Y. Suryawanshi, K. Patil, and P. Chumchu, "Coconut (*Cocos nucifera*) tree disease dataset: A dataset for disease detection and classification for machine learning applications," *Data Brief*, vol. 51, Dec. 2023, Art. no. 109690.



international Journals and conferences attend more than 30 Workshops/FDPs/Seminars. Her areas of interest are image processing and machine learning.



CMR UNIVERSITY, Bengaluru, Karnataka, India. He has 21 years of teaching and research experience. His areas of interest are AI, machine learning, deep learning, and NLP.

L. K. Sowmya Sundari received the A.M.I.E. (CSE) degree from The Institution of Engineers (India), in 2012, and the M.Tech. degree from Visvesvaraya Technological University (VTU), Belgaum, India, in 2015.

She is an Assistant Professor with the School of Computing and Information Technology, REVA University, Bengaluru. She is having ten years of teaching experience in engineering colleges. She has 14 research publications to her credit, published in various national and international Journals and conferences attend more than 30 Workshops/FDPs/Seminars. Her areas of interest are image processing and machine learning.

Mallikarjun M. Kodabagi received the B.E. degree in computer science and engineering from Basaveshwar Engineering College, Bagalkot, Karnataka, India, the M.Tech. degree in computer science and engineering from the University of Mysore, Mysore, India, and the Ph.D. degree from the Department of Computer Science and Engineering, Visvesvaraya Technological University, Belgaum, Karnataka, India.

He is working as a Professor with the School of Engineering and Technology, Director IQAC, CMR UNIVERSITY, Bengaluru, Karnataka, India. He has 21 years of teaching and research experience. His areas of interest are AI, machine learning, deep learning, and NLP.



B. Muthu Kumar is a Professor with the School of Computing and Information Technology, REVA University, Bengaluru, India. He supervised four research scholars at Anna University, Chennai, India, and currently guides four research scholars in REVA University. His research interests include image processing, wireless networks, the Internet of Things (IoT), and computing techniques.



BalaAnand Muthu received the Ph.D. degree from Anna University, Chennai, India, in 2019.

He currently holds the position of an Associate Professor with the Department of Computer Science and Engineering, Tagore Institute of Engineering and Technology, Salem, Tamil Nadu, India.



Mohammed Altaf Ahmed was born in India. He received the bachelor's degree in electronics and communication engineering from Swami Ramanand Teerth Marathwada University, Nanded, India, in 2000, the master's degree in embedded systems from Jawaharlal Nehru Technological University, Hyderabad, India, in 2008, and the Ph.D. degree in engineering from GITAM University, Vishakhapatnam, India, in 2018.

He is an accomplished academic and a Researcher specializing in VLSI design, embedded systems, IoT devices, and AI applications in engineering. He is currently an Assistant Professor with the Computer Engineering Department, Prince Sattam Bin Abdulaziz University (PSAU), Al-Kharj, Saudi Arabia. In addition to his teaching role, he is actively engaged in academic development, serving as a Coordinator in the Quality and Development Unit of the department. He has successfully led and contributed to multiple research grant projects in his field and has published research articles in leading academic journals indexed in Scopus and Science Citation databases. His contributions continue to shape advancements in embedded systems, AI-driven engineering solutions, and VLSI design.

Dr. Ahmed received the prestigious Best Thesis Award and the Gold Medal for his research excellence in 2018.



A. Ahilan received the Ph.D. degree from Anna University, Chennai, India, in 2017.

He is working as an Associate Professor with the Department of Electronics and Communication Engineering, PSN College of Engineering and Technology, Tirunelveli, India. His research interests include FPGA prototyping, computer vision, the Internet of Things (IoT), and cloud computing in medical and biometrics.